PRACTICAL ASSIGNMENT 4: Evidence Preservation

1. Objective

The objective of this task is to practice digital evidence preservation by collecting volatile data and acquiring memory evidence using forensic tools. Additionally, the task ensures proper chain-of-custody documentation and integrity verification using hashing techniques.

2. Tools Used

- Velociraptor
- Windows PowerShell

3. Environment Setup

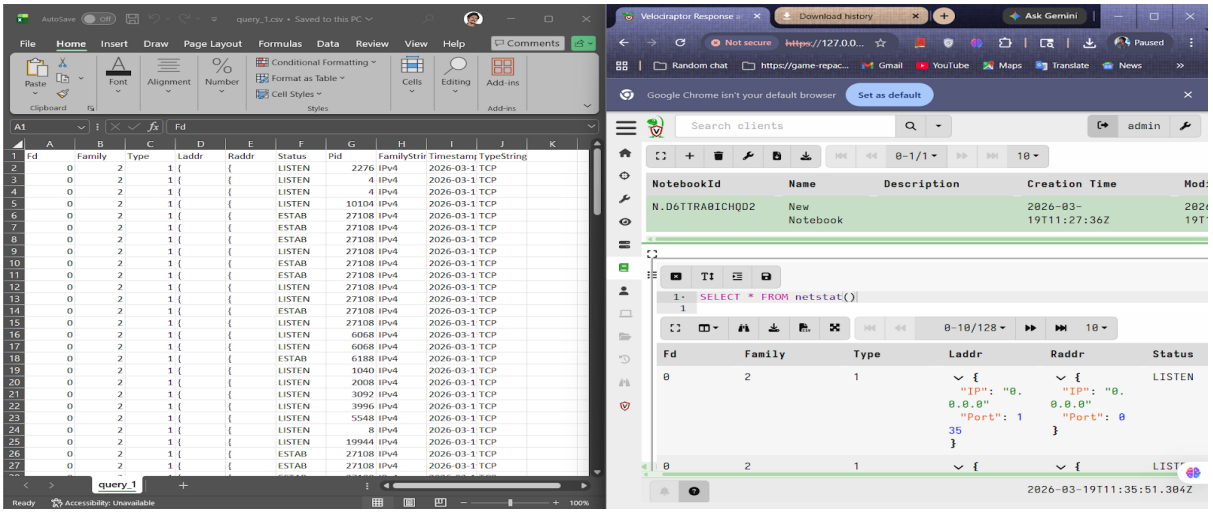
- Host Machine: Windows 11
- Tool Used: Velociraptor (Local Server + Client Mode)
- Target System: Same machine (THIS_PC)

4. Volatile Data Collection

Volatile data refers to temporary system data that is lost when the system is powered off. In this task, network connection data was collected using Velociraptor.

Steps Performed

1. Opened Velociraptor Web Interface
2. Navigated to **Hunt Artifact**
3. Selected artifact related to system/network monitoring
4. Executed query: `SELECT * FROM netstat`
5. Data was successfully collected and exported as a CSV file.

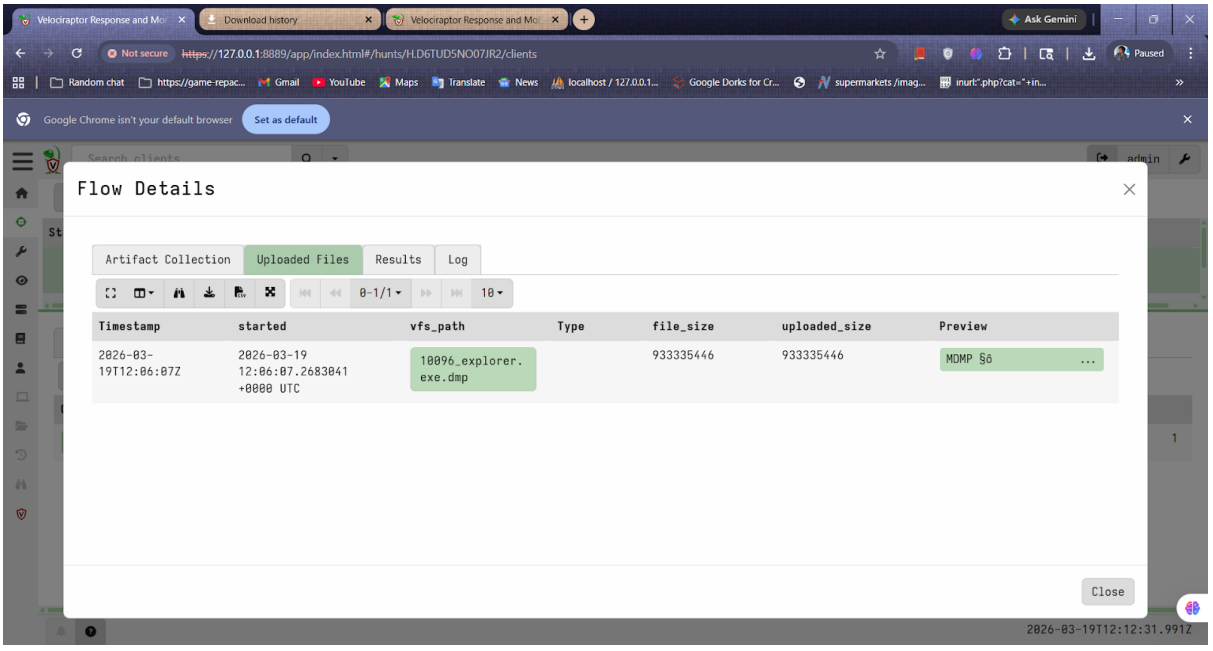


5. Evidence Collection (Memory Dump)

Memory acquisition was performed using Velociraptor to collect process-level memory evidence.

Steps Performed

1. Navigated to **Hunt Artifact**
2. Selected: `Windows.Memory.ProcessDump`
3. Configured parameters:
   - ProcessRegex: `explorer.exe`
4. Launched the hunt
5. Memory dump was successfully generated



6. Evidence Integrity Verification

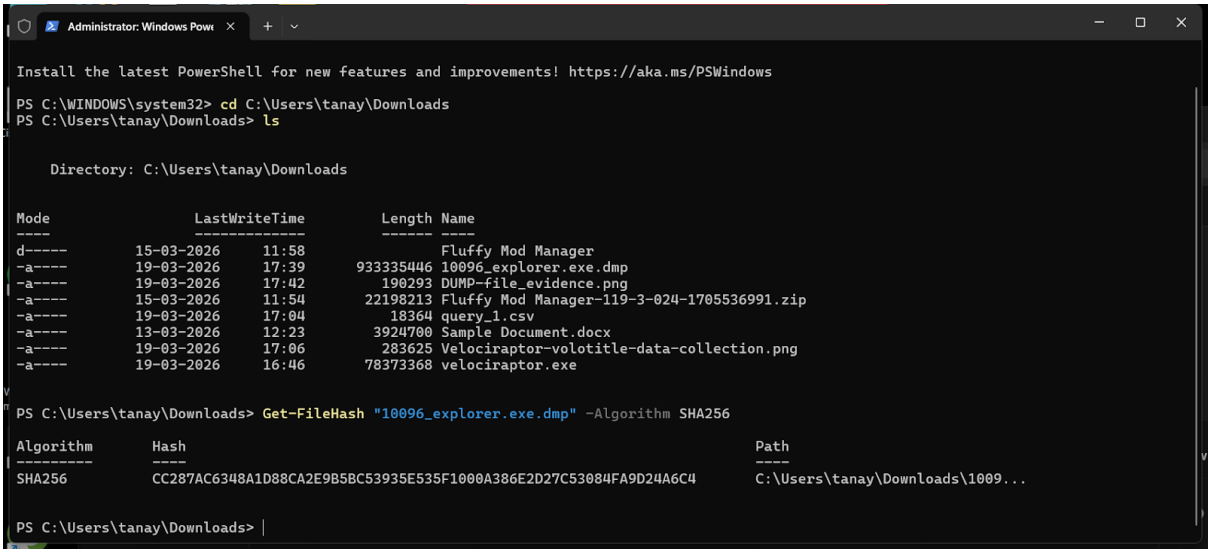
To ensure forensic integrity, the memory dump was hashed using SHA256.

Command Used

```
Get-FileHash "10096_explorer.exe.dmp" -Algorithm SHA256
```

Generated Hash

```
CC287AC6348A1D88CA2E9B5BC53935E535F1000A386E2D27C53084FA9D24A6C4
```



7. Evidence Documentation

Item	Description	Collected By	Date	Hash Value
Memory Dump	explorer.exe process dump	SOC Analyst	2026-03-19	CC287AC6348A1D88CA2E9B5BC53935E535F1000A386E2D27C53084FA9D24A6C4

8. Conclusion

The task was successfully completed by:

- Collecting volatile network data
- Acquiring process memory dump
- Verifying integrity using SHA256 hashing
- Documenting evidence in a structured format